

Honors Option Final Report

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For my Honors project, I developed a modular Benchmarking Suite to analyze the scalability of the Company B campus simulation. Beyond my coursework on the Group 12 Swarming Agents, I built a "plug-in" benchmarking tool using the Registry Design Pattern. By utilizing `std::function` and `lambdas`, I created a method that allows any developer to register simulation logic for testing without modifying their source code. To get to this point, I researched how to design a plug-in module, which taught me how benchmarking tools are built and how to make the process as easy as possible for other developers.

A majority of my time was spent researching system-level resource tracking. I implemented `getrusage` to query the OS kernel for Resident Set Size (RSS), which taught me how hardware manages memory allocation at a deeper level than standard C++. This allowed me to provide users with data on memory "deltas" and heap growth, ensuring our simulation could scale to thousands of agents without crashing the integrated environment.

I also researched how to eliminate "noise" in performance data. This noise can pollute data to the point where it fails to capture the true metrics of the desired functionality. I implemented CPU warmup cycles and a headless execution mode to ensure that data wasn't skewed by cold caches or GPU overhead. This taught me that we cannot simply time a feature once and call it a day; instead, we must warm up the CPU so that other system-related factors do not impact our metrics.

Using this Benchmarking Suite, we can also conduct scientific experiments on hyperparameters, such as agent count, seed, repetitions, etc. This allows us to see which parameter combinations might cause the simulation not to scale properly or identify where performance bottlenecks occur.

Finally, I learned how to integrate this benchmarking tool into an existing workflow. I added a "make" command to help run benchmarking scripts written by any team. I also ensured that the benchmarking suite added zero overhead to the production build. This project was very new and fun for me to work on since I was able to work with concepts I hadn't explored before and see firsthand how this tool is useful to the project.

Video: <https://youtu.be/L1vNm2CRB0Y>